PROGRAMMING IN JAVA

Assignment 11

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

How do you establish a connection to a database using JDBC?

a. **By creating an instance of the Connection interface**

b. **By using the DriverManager.getConnection() method**

c. **By implementing the Connection interface**

d. **By extending the Connection class**

Correct Answer: b

b. **By using the DriverManager.getConnection() method**

Detailed Solution:

To establish a connection to a database using JDBC, you use the DriverManager.getConnection() method. This method takes a JDBC URL, username, and password as parameters and returns a Connection object, which represents a connection to the database. The JDBC URL specifies the database type, location, and other connection details.

QUESTION 2:

Which method executes a simple query and returns a single Result Set object?

a. **executeQuery()**

b. **executeUpdate()**

c. **execute()**

d. **run()**

Correct Answer: a

a. **executeQuery()**

Detailed Solution:

The executeQuery() method is used to execute a simple SQL query that returns a single ResultSet object.

QUESTION 3:

Which of the following ensures that the correct driver is used to access each data source?

a. **java.sql.Manager**

b. **java.sql.DriverManager**

c. **java.sql.Statement**

d. **java.sql.DriverSession**

Correct Answer: b

b. **java.sql.DriverManager**

Detailed Solution:

The DriverManager class ensures that the correct driver is used to access each data source. It manages a list of database drivers and handles establishing connections to the database.

QUESTION 4:

What is the purpose of the ResultSet interface in JDBC?

a. **To store the result of a query**

b. **To execute SQL queries**

c. **To manage database connections**

d. **To update data in the database**

Correct Answer: a

a. **To store the result of a query**

Detailed Solution:

The ResultSet interface in JDBC represents the result set returned by a query. It provides methods to navigate through the result set, retrieve data, and get information about the columns in the result set. The ResultSet interface allows you to process and extract data from a database query.

QUESTION 5:

What does setAutoCommit(false) do?

a. **commits transaction after each query**

b. **explicitly commits transaction**

c. **does not commit transaction automatically after each query**

d. **never commits transaction**

Correct Answer: c

c. **does not commit transaction automatically after each query**

Detailed Solution:

setAutoCommit(false) does not commit transaction automatically after each query. That saves a lot of time of the execution and hence improves performance.

QUESTION 6:

The following is a statement in Java using JDBC.

```
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/nptel","joy","java"
    );
```

Which of the following statement is FALSE?

a. **3306 is the default MySQL port.**

b. **Database name is 'nptel'**

c. **The database server is hosted on IP 127.0.0.1**

d. **Username is 'java' and the password is 'joy'**

Correct Answer: d

d. **Username is 'java' and the password is 'joy'**

Detailed Solution:

In the statement, joy is the username and java is the password. Therefore, statement d, which incorrectly claims that the password for the java user is joy, is false.

QUESTION 7:

What is the correct order to close database resources?

a. **Connection then Statement then ResultSet**

b. **ResultSet then Statement then Connection**

c. **Statement then Connection then ResultSet**

d. **Statement then ResultSet then Connection**

Correct Answer: b

b. **ResultSet then Statement then Connection**

Detailed Solution:

The correct order to close database resources is to first close the ResultSet, then the Statement, and finally the Connection. This ensures that all resources are properly released in the correct sequence.

QUESTION 8:

What is the full form of JDBC?

a. **Java Database Connectivity**

b. **Java Data Code**

c. **Java Data Communication**

d. **Java Development Connectivity**

Correct Answer: a

a. **Java Database Connectivity**

Detailed Solution:

JDBC stands for Java Database Connectivity, a Java API used to connect and interact with relational databases. It provides methods to query and update data in a database.

QUESTION 9:

Fill in the missing code to establish a connection to a MySQL database.

```
import java.sql.*;

public class JDBCExample {
    public static void main(String[] args) {
        try {
            // Load JDBC Driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Establish Connection
            Connection connection = // INSERT CODE HERE;

            System.out.println("Connected to the database.");
            connection.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

What should replace `// INSERT CODE HERE` ?

a. **DriverManager.connect("mysql:localhost:mydb", "user", "password");**

b. **DriverManager.getConnection("jdbc:mysql://localhost:3306/mydb", "user", "password");**

c. **Connection.get("jdbc:mysql://localhost:3306/mydb", "user", "password");**

d. **Driver.connect("jdbc:mysql://localhost:mydb", "user", "password");**

Correct Answer: b

b. **DriverManager.getConnection("jdbc:mysql://localhost:3306/mydb", "user", "password");**

Detailed Solution:

The `DriverManager.getConnection` method establishes a connection to the specified database URL with the provided username and password.

QUESTION 10:

What is the purpose of the `DriverManager` class in JDBC?

a. **To manage database connections.**

b. **To execute SQL queries.**

c. **To fetch data from a database.**

d. **To represent a database record.**

Correct Answer: a

a. **To manage database connections.**

Detailed Solution:

The `DriverManager` class loads the JDBC drivers and establishes connections to databases.